

Detection of Parkinsonian syndrome using a wearable device

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Abstract

Parkinson's disease is a brain disorder that impairs the central nervous system. Current methods of testing the symptoms of Parkinson's disease are either expensive or extremely sophisticated for people to handle. To address these shortcomings, a novel system is designed which is portable, wearable, accessible, economical, and easy to use. The designed system is capable of analyzing the hand tremors and Parkinsonian gait of Parkinson's disease symptoms. To acquire these symptoms, an EMG Sensor, an MPU6050 GY291, and an IMU accelerometer are connected to the Arduino Uno board. The results indicate that hand tremors and gait parameters can be effectively detected with the designed system. The proposed system design also includes the facility to send the results to a consulting physician via email. Thus, this system has the potential to improve diagnosis, treatment, and research in the field of testing patients suffering from this disease.

Keywords: Parkinson's disease, gait, hand tremors, EMG, MPU6050 GY291, IMU.

1 Introduction

Parkinson's disease, mostly found in older people, is one of the chronic neurodegenerative diseases which exhibits a progressive loss of the ability to coordinate movements [1]. Symptoms of Parkinson's disease that can be detrimental to the quality of life of patients and are often disabling include, bradykinesia, resting tremor, depression and cognitive impairment, rigidity, and postural instability [2]. Out of these, the two main symptoms of Parkinson's disease are Parkinsonian Gait and hand tremors. There is an immediate requirement in the medical field for early detection and monitoring of Parkinson's disease and there is a need for the development of diagnostic tools to characterize progressive movement disorders. It is important to test hand tremors and at the same time perform gait analysis. This has led to the implementation of systems that can monitor the symptoms in a non-controlled environment [3]. In recent years, due to technological advancements, mankind has developed different types of systems to test Parkinson's Disease. There have been a lot of systems that were developed as analyzed in [4], [5], [6], [7], [10], and [11] but most of them turned out to be either inaccessible or expensive. Patients may not be comfortable in using huge machines with sophisticated features especially when it is known that the largest risk factor is the age of an individual for the development and progression of this disease [8]. Hence, a simple and easy-to-use system is proposed in this research paper which can test the presence of Gait and Tremors within minutes. Further, these outputs can be used by physicians to provide a further diagnosis. These outputs can also be uploaded on the cloud and auto-emailed to consulting physicians if the patient desires. This system is not only cost-effective and comprehensive but also is designed in a way that people of all age groups can handle the wearable device comfortably. In addition to this introduction, the novel system designed and developed in this work is explained in section-2, results, and discussion is presented in section-3, and conclusions are given in section-4.

2 Proposed System Design

A wearable system was developed for testing Parkinsonian Gait and tremors in hands. Patients of this disease can use this testing system with or without the help of an examiner. The system is divided into two parts: the Hand Tremor Testing system and the Gait Analysis system. This is done to cover both the symptoms in one device itself. For the Hand Tremor Testing system, we have used two methods to test an EMG sensor and an MPU6050 GY291. The output of EMG Sensor will be sensor value and mV whereas for MPU6050 GY291 we will get accelerations in all three axes. For the Gait Analysis System, various Gait parameters such as Yaw, Pitch, Roll, Euler angles, acceleration, and real-world acceleration are obtained. These gait parameters are recorded by attaching the system to

the ankle, while the patient walks for a distance of 4.5 meters. An Arduino open-source prototyping platform is used where results would be displayed. These results can be sent to a physician via email server through the examiner and the details are shown in Figure-1.

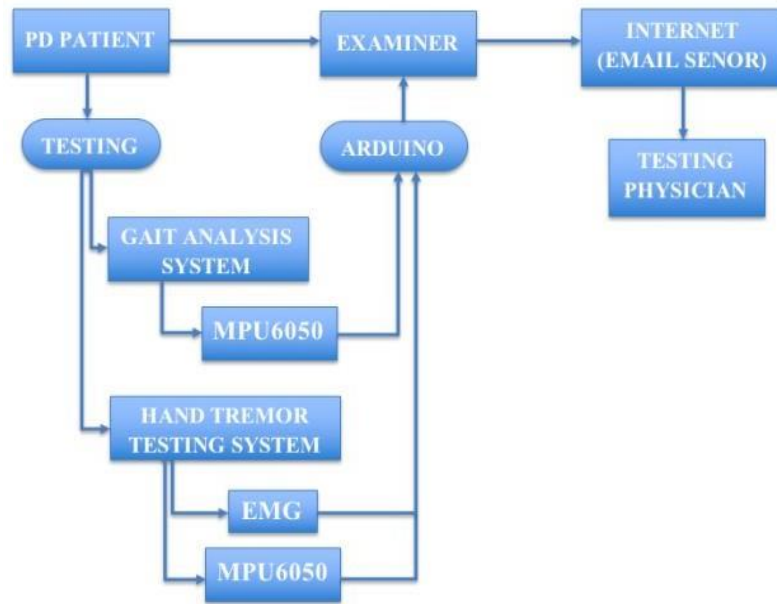


Fig. 1. Block Diagram of proposed system

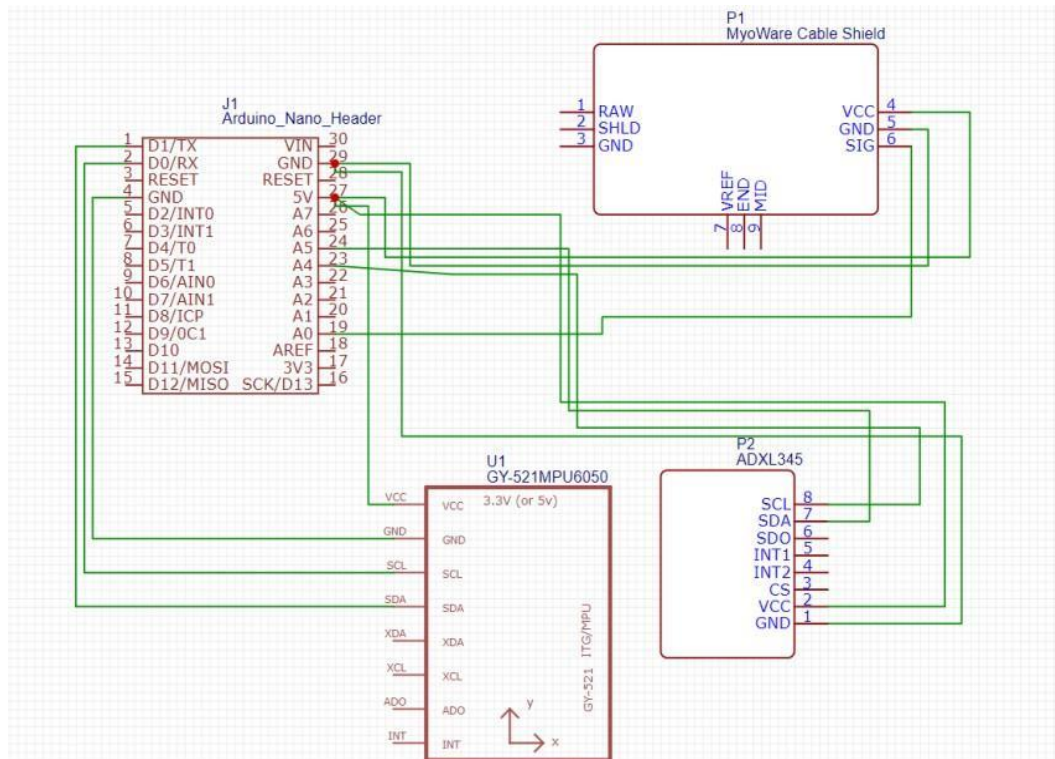


Fig. 2. Schematic Diagram of proposed system

2.1 Hand Tremor Testing with EMG

A system made up of EMG Sensor, Arduino Nano, Biomedical sensor pads, connecting wires and the USB cable is used for the detection of tremors. The system would be connected to the patient's hand and based on contractions and relaxations; a graph is plotted in a serial plotter of Arduino IDE. These graphs can be mailed to your doctor for proper diagnosis as well. An EMG detects when muscle cells are neurologically or electrically activated, from the electrical potential generated by them. These signals can be analyzed to detect activation levels, biomechanics of human body movement, medical abnormalities, and recruitment order [8].

2.2 Hand Tremor Testing with MPU

The second testing system is made up of MPU6050 GY291, Arduino Nano, connecting wires, and USB cable. This system would be connected to the finger of the patient through a glove. The system would provide us with accelerations for all three axes. Then using equation (1), motion frequency and amplitude of tremors for all the three axes are computed.

$$S_f = S_o + \frac{(AT)^2}{2} \dots (1)$$

where S_f is final displacement and S_o is initial displacement = 0.

2.3 Gait Analysis System

The Gait Analysis system consists of an Arduino Nano and an MPU6050 sensor (three-axis accelerometer and three-axis gyroscope), which is attached to the ankle of the patient with the help of a band. This is used for acceleration sensing, calculating the pitch angle of the leg's swing, and transmitting real-time signals. This can be used for finding the difference between a Parkinson's disease patient and a normal individual, and can also be processed for further quantitative studies. The study of human locomotion is done using gait cycle analysis. Some gait features are specific to those with Parkinson's disease and get worse with time. An objective and quantitative gait analysis system can aid in diagnosis and improve the current methods in risk assessment and prevention [9].

3 Results and Discussion

3.1 Hand Tremors

3.1.1 EMG Sensor

The proposed hardware of the system which is used for testing hand tremors is shown in Figure-2. As soon as the biomedical sensor pads are placed and connections are made in the entire system, a graph is obtained on the serial plotter. The graph shown in Figure-3 explains variation based on the movement of the hand. When there is little to no movement, approximately a straight line is obtained. However, if the system detects tremors i.e., abrupt movements, the graph rises, as is shown after a certain point in the graph given below in Figure-3. The obtained values of sensor values and mV values on Serial Monitor are shown in Figure-4



Fig. 3. Hardware of the designed system for testing tremors



Fig. 4. Graph of value obtained and mV used when a test run was conducted on the entire system

```

COM3
22:50:34.289 -> Sensor Value: 227.00
22:50:34.330 -> Voltage: 1109.48 mV
22:50:34.330 ->
22:50:34.330 -> Sensor Value: 227.00
22:50:34.365 -> Voltage: 1109.48 mV
22:50:34.407 ->
22:50:34.407 -> Sensor Value: 226.00
22:50:34.407 -> Voltage: 1104.59 mV
22:50:34.407 ->
22:50:34.407 -> Sensor Value: 227.00
22:50:34.446 -> Voltage: 1109.48 mV
22:50:34.489 ->
22:50:34.489 -> Sensor Value: 227.00
22:50:34.489 -> Voltage: 1109.48 mV
22:50:34.529 ->

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Fig. 5. Sensor value and mV value as displayed on Serial Monitor

3.1.2 MPU6050 GY291-in Hand Tremor Detection

In this setup, Arduino Nano is connected to MPU6050 GY291 which would be inside a hand glove. Hence, the patient has to wear these gloves which results in the system running and values of accelerations are obtained. These are fed into the code to get displacement values and FFT. From displacement values, the values of frequency of Tremors can be calculated which can be sent to the Physician for further diagnosis.

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COM6
Acceleration X: -4.90 , Y: -0.78 , Z: 9.49 m/s^2
Acceleration X: -2.24 , Y: -1.10 , Z: 8.32 m/s^2
Acceleration X: -0.98 , Y: -1.14 , Z: 8.39 m/s^2
Acceleration X: -2.20 , Y: -0.82 , Z: 9.14 m/s^2
Acceleration X: -2.43 , Y: -1.37 , Z: 9.41 m/s^2
Acceleration X: -2.12 , Y: -1.69 , Z: 8.75 m/s^2
Acceleration X: -2.24 , Y: -1.26 , Z: 9.18 m/s^2
Acceleration X: -2.35 , Y: -0.35 , Z: 9.30 m/s^2
Acceleration X: 0.12 , Y: 0.63 , Z: 10.79 m/s^2
Acceleration X: -2.20 , Y: -1.02 , Z: 9.65 m/s^2
Acceleration X: -2.12 , Y: -1.96 , Z: 8.43 m/s^2
Acceleration X: -1.88 , Y: -1.29 , Z: 8.98 m/s^2
Acceleration X: -1.80 , Y: -1.29 , Z: 8.94 m/s^2
Acceleration X: -1.88 , Y: -1.33 , Z: 9.02 m/s^2
Acceleration X: -1.88 , Y: -1.37 , Z: 8.87 m/s^2

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Fig. 6. Acceleration values in all three axes as displayed on Serial Monitor

3.1.3 Testing Values

HAND TREMOR		SENSOR NAME			
VALUE NUMBER	EMG		MPU 6050		
	Sensor Value	mV	x(m/s^2)	y(m/s^2)	z(m/s^2)
1	227	1109.48	-1.06	-0.27	9.22
2	227	1109.48	-1.1	0.31	8.51
3	226	1104.59	-2.12	0.16	9.89
4	227	1109.4	-2.24	-0.63	19.22
5	227	1109.48	0.51	0.43	4.67
6	226	1104.59	-0.98	0.24	7.88
7	226	1104.59	-1.45	1.06	9.1
8	226	1104.59	-2.28	-0.98	9.49
9	225	1099.71	-3.22	-1.41	13.38
10	227	1108.48	-2.12	2.2	7.96
AVERAGE	226.4	1106.439	-1.606	0.111	9.932

Fig. 7. Values for Hand Tremors when tested on healthy Patient A

3.2 Gait Analysis

The proposed prototype of the system consists of Arduino Nano and MPU6050, with which the following Gait parameters can be obtained: Euler angle, Pitch, Yaw, Roll, Quaternion values, Acceleration, and World-frame Acceleration values. The system is attached to the leg, on the ankle and the patient is made to walk a distance of 4.5 meters for analysis purposes. The recorded values are then used to detect the Parkinsonian Gait and can be further processed into a quantitative study which would help detect Parkinson's Disease at an early stage.

3.2.1 Euler angles and Acceleration

Figure-8 shows the tabulation of Euler angles and acceleration. The Euler angles portraying the orientation of a moving frame of the individual's motion are used for the pitch, yaw, and roll angles. Acceleration of the subject is obtained by an accelerometer.

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Time,Pitch,Yaw,Roll
123.38,8.21,179.45
123.43,8.22,179.46
123.47,8.21,179.47
123.52,8.21,179.47
123.57,8.22,179.48
123.62,8.21,179.49
123.67,8.21,179.49
123.72,8.21,179.51

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Fig. 8. Euler angles and acceleration values displayed on serial monitor

3.2.2 Yaw, Pitch, Roll, and Quaternion

Figure-9 shows the tabulation of yaw, pitch, roll, and quaternion values. A lot of fluctuation is observed in yaw and roll and it fully deviates from the particular degree once the pitch angle reaches 90°. Thus, to avoid singularity, it is suggested to avoid axis-exchanged compensation factoring quaternions.

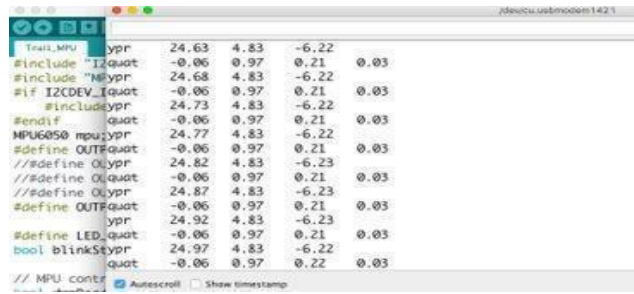


Fig. 9. Yaw, Pitch, Roll and Quaternion values displayed on serial monitor

3.2.3 World Acceleration

Figure-10 shows the tabulation of world acceleration. In this, world acceleration displays initial world-frame acceleration, adjusted to remove gravity and rotated based on known orientation from known Quaternion values.

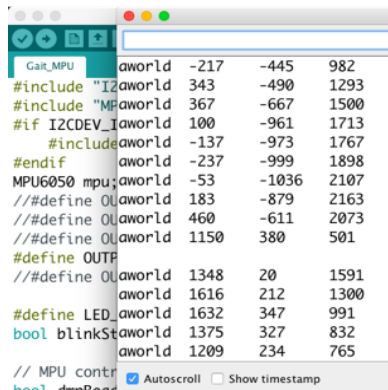


Fig. 10. World acceleration values displayed on serial monitor

3.2.4 Output Waveforms

The following output waveforms are obtained from Arduino's Serial Plotter, where the results of a healthy person are compared with a patient who could be potentially affected with Parkinson's. The following outputs are recorded with a healthy person walking in a straight line of 5m, while the patient walks in a straight line of half the distance, i.e., 2.5m, due to their reduced speed.

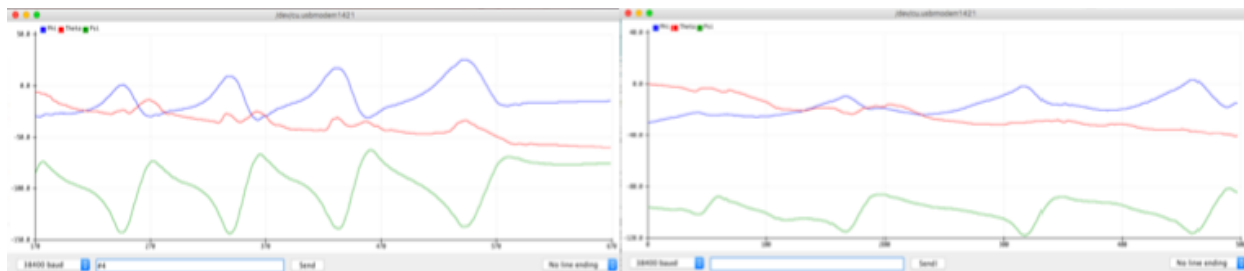


Fig. 11. Results of a Healthy Person vs a Patient – Euler Angles (Phi, Theta, Psi)

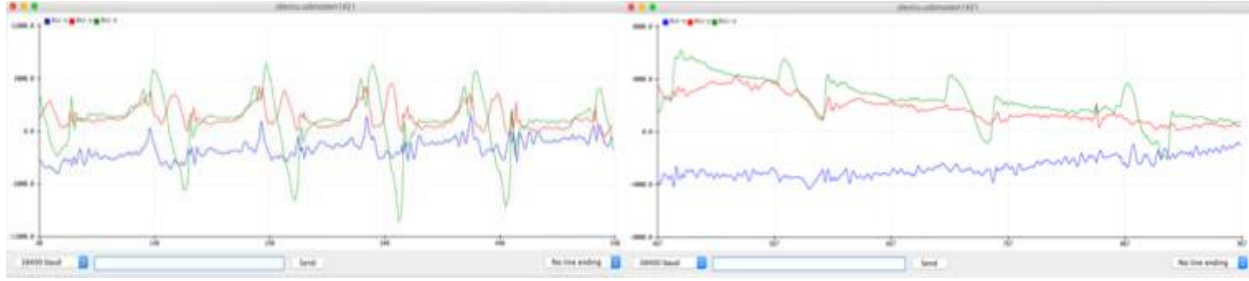


Fig. 12. Results of a Healthy Person vs a Patient – Acc. with respect to x, y, z - axes

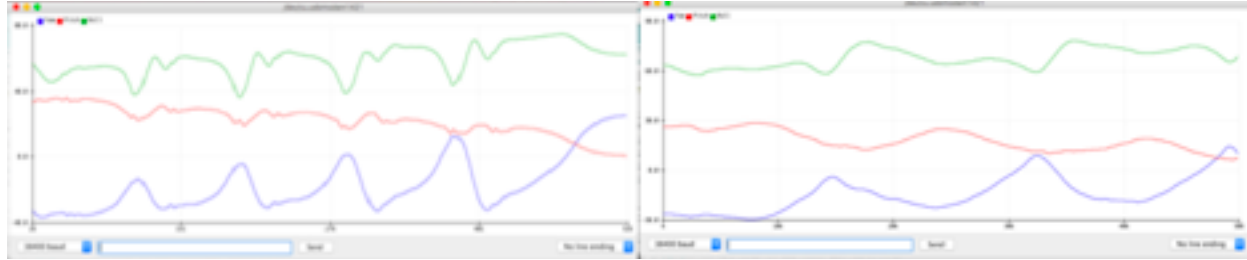


Fig. 13. Results of a Healthy Person vs a Patient – Yaw, Pitch, Roll

3.3 Discussion

For Parkinson's Disease, two tests were successfully studied to record observations. The principal advantage of both tests is that they provide prominent results and can bridge the gap between new technology and the current clinical testing processes. The graph of Hand Tremors is more perceptible and better visualized when the sensor is placed on the muscle (arm), whereas the Gait is measured with the sensor placed on the ankle.

4 Conclusion

This paper presents an approach for the comprehensive development of a novel system to measure Parkinson's Disease tremors and gait analysis. The objective of this research is to develop economical and effective solutions based on simple techniques while ensuring personal comfort through the use of mountable sensors such as gyro sensors and EMG sensors. The chosen sensor for gait analysis proves to be effective for acquiring the necessary data. Various gait parameters such as stride time, stance time, step time, swing time, step length, gait speed, and stride length can be extracted from the developed system using the IMU accelerometer and gyroscope. When compared to persons with certain deformities, uniform repetitive flexion and extension of joints are observed at standard gait, which is a significant result of the proposed system. An appropriate methodology of analysis can be designed to serve as an appropriate rehabilitation process for patients with gait malfunctions. To facilitate optimal management during the monitoring, a user-friendly interface can help to send the data to the respective physicians.

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